

Ibrahim Al Azher

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EXPERIENCE

Research Assistant

May 2024 - August 2024

Northern Illinois University

Dekalb, IL

- Extracting texts from scientific articles using API. Incorporating LLM techniques such as zero-shot, few-shot, chain of thoughts, self-consistency, self-improvement, and fine-tuning with Retrieval Augmented Generation (RAG) for text generation.
- LLM for code vulnerability detection, image analysis with large vision language models
- Collaboratively written NSF Grant proposal.

Teaching Assistant

August 2024 - Present, Aug 2023 - May 2024

Northern Illinois University

Dekalb, IL

- C++, Data Structures and Algorithm Analysis. Conducted weekly lab sessions for over 39 students, facilitating hands-on exercises in Data structures and Algorithms.

Software Engineer

Dec 2020 - May 2021, June 2021 - Feb 2022

Unisoft Systems Limited, BYSL Global Technology Group

Dhaka, Bangladesh

- Works on scalable projects from design to coding, testing, and installation using Python, XML, Javascript, PostgreSQL, and jQuery in Linux OS.
- Linux OS. Designed and developed various ERP-based business software and modules like sales, inventory, purchase, website, and e-commerce using Python language in Odoo.

PUBLICATIONS

Published

- “Identifying Author in Bengali Literature by Bi-LSTM with Attention Mechanism.” 24th International Conference on Computer and Information Technology (ICCIT), December 2021.
<https://doi.org/10.1109/ICCIT54785.2021.9689840>. In this paper, we propose a Bi-LSTM model with self-attention mechanism to address Authorship Attribution problem. Use GloVe for embedding. We evaluate our model on multiple datasets and experiment with various architectures such as CNN, BiLSTM, GRU, FastText, CNN + BiLSTM. This work published as a full paper.
- “LimTopic: LLM based Topic Modeling and Text Summarization for Analyzing Scientific Articles Limitations.” 2024 ACM/IEEE Joint Conference on Digital Libraries, Hong Kong. We extracted the scientific article’s limitations sections using a parsing tool. We introduce LimTopic, a strategy for using topic generation in limitation sections in scientific articles with large language models (LLMs). We applied prompt engineering and fine-tuning approaches, experimented with various LLMs (GPT 3.5, GPT 4, and Llama 2) integrated with BERTopic, and found that BERTopic with GPT 4 outperforms. To summarize each topic, we applied various LLMs as text summarizers. [Code](#). This work published as a full paper.
- “Limitation Generation of Research Papers with LLMs and Retrieval Augmented Generation (RAG).” The 11th IEEE International Conference on Data Science and Advanced Analytics (DSAA 2024). Considering various sections (Abstract, Introduction, Methodology, and so on) from scholarly articles and generating Limitation sections using various LLMs such as BART, T5, Pegasus, GPT 3.5, GPT 4, and Llama. We implemented the cosine similarity approach to find important sections of a paper, integrated RAG with GPT 3.5, and performed better in the score. This work published as a poster. In an extended version of this work, we are incorporating ‘OpenReview’ with the paper’s limitations as an additional ground truth. We are making more interpretable evaluation metrics by pointwise similarity between ground truth and generated limitations other than BERTScore, using LLM as feedback for self improvement, and feedback used for better prompt engineering and RAG.
- “Mitigating Visual Limitations of Research Papers.” 2024 IEEE International Conference on Big Data (IEEE BigData 2024). We focus on visual limitations in scientific research papers such as charts, graphs, and diagrams. Our task is to build a multimodal LLM to generate clear, concise descriptions, regenerate visual images, and explainable tabular forms based on images. We are integrating Vision Transformer with LLM to improve poor visualizations, incomplete data representations, and misaligned captions. This work published as a poster.

In Progress

- “Can ChatGPT predict citations of Research Article?”. Take various domains of scientific articles and consider section abstract, introduction, and other important sections, as well as some other metrics such as readability, and experiment with GPT 3.5 and GPT-4o to predict the citation using prompt engineering technique. We are also experimenting with images from scientific articles using the Vision Language model (VLM).
- “Limitation Recommendation with Graph Neural Network and/or LLM.” The graph-based method involves constructing a limitation recommendation system using a graph neural network to represent relationships among scientific articles based on their limitations. Each node represents a limitation section, title, topics, and other sections from a paper. Creating a knowledge graph using citation of papers building edges will suggest a new paper’s additional limitations based on the knowledge graph with Graph Neural Network and Attention. In addition, we are incorporating graph to LLM by i) sending subgraph to , ii) sending the textual data to LLM (recommendation) to generate refine limitations.
- “How Large Language Models (LLM) affects the Readability score in Scientific article.” Here, we are taking the scientific papers from before the LLM era and after and measuring the readability score to find how these metrics affect various disciplines.
- “Interpretable Topic Modeling” It is the extended version of LimTopic work where we find the correlation between each topic with input embedding and which input categories are liable to make certain topics.
- “Predict the next scientific article’s content using time series analysis.” Here, we are utilizing time series analysis with the publication of research articles in certain categories to predict the trend by predicting the types of research that will be published soon.

Machine Learning works | *Python, Keras, Tensorflow*

Jan 2020 – July 2022

- Predicting mortality, hospital length of stay in health dataset (MIMIC IV). Analysis Amazon review datasets with various models with various data preprocessing. Apply CNN, LeNet, AlexNet, GoogleNet, ResNet, VGG16, VGG19 model as well as transfer learning model on image data.

EDUCATION

Northern Illinois University

PhD Computer Science; GPA: 3.67

Dekalb, IL

August 2023 - present

Shahjalal University of Science and Technology

B.Sc. Computer Science; GPA: 3.69

Sylhet, Bangladesh

March 2016 - March 2021

SKILLS

Programming Languages: C, C++, Python, Javascript, Java, PHP

Framework and Technologies: Django, Laravel, HTML, CSS, Bootstrap, Odoo, Django REST framework

Open Source Library: Keras, Tensorflow, Pytorch, numpy, pandas, scikit learn, matplotlib,

(Tools): Oracle Sql, MySQL, Postgresql, Git, Github, CodeBlocks, Pycharm, Visual Studio, Anaconda, Jupyter notebook, Latex, Overleaf

HONORS & AWARDS

- **Google GCP Grant Award**, Google — Feb 2024-Feb 2025
- **Travel Grant, Volunteer**, IEEE International Conference on Data Science and Advanced Analytics, San Diego, California - October 2024
- **SIGIR student Travel Grant**, 2024 ACM/IEEE Joint Conference on Digital Libraries (JCDL), Hong Kong - Decemeber 2024

AREA OF EXPERTISE

My research area is machine learning, information extraction, natural language processing, and large language models (LLM). Our research goal is to make scientific articles more reliable, credible, robust, reproducible, and transparent to the research community and policymakers. My work also incorporates LLM techniques such as zero-shot, few-shot, chain of thoughts, self-consistency, and fine-tuning. I am also investigating reducing the transformer’s quadratic time to linear time using Mamba, flash attention, the KV cache approach, xLSTM, and the sparse LLM process. I’m also working on LLM for code vulnerability detection and image analysis with large vision language models, superpixels, diffusion models, vision transformers, etc. I have strong background in both machine learning and deep learning models, as well as in the mathematical analysis of these models using statistics, probability, and linear algebra.